UNIVERSITÀ DEGLI STUDI DI UDINE

Laboratorio di Programmazione Web
(laurea in Scienze e Tecnologie Multimediali)

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Web link alle pagine del corso:

http://elearning.uniud.it/moodle/course/view.php?id=6288

Server side programming
PHP scripting language

I materiali in queste slide sono tratti in parte da:

V. Della Mea, L. Di Gaspero, I. Scagnetto,"Programmazione lato server", Apogeo.
http://www.w3.org/
http://en.wikipedia.org (Wikipedia)
http://www.php.net

Operations and
functions on strings

The PHP language syntax

I. Operations and functions on strings

- A string can be seen as an array of characters.

- Characters within strings may be accessed and modified by specifying the zero-based offset of the desired character after the string using square array brackets:

```php
<?php
 // Get the first character of a string
 $str = 'This is a test.';
 $first = $str[0];

 // Get the third character of a string
 $third = $str[2];

 // Get the last character of a string.
 $str = 'This is still a test.';
 $last = $str[strlen($str)-1];

 // Modify the last character of a string
 $str = 'Look at the sea';
 $str[strlen($str)-1] = 'e';
?>
```

The PHP language syntax

I. Operations and functions on strings

- The ***strlen()*** function returns the length of the given string (0 if the string is empty).

```php
<?php
  $str = 'abcdef';
  echo strlen($str); // 6

  $str = ' ab cd ';
  echo strlen($str); // 7
?>
```

The PHP language syntax

I. Operations and functions on strings

- PHP provides a large number of built-in functions dedicated to strings

- Among the most useful: ***manipulation*** of strings, e.g.:
  - ***trim*** : strip whitespace (or other characters) from the beginning and end of a string
  - ***substr*** : return part of a string
  - ***str_replace*** : replace all search string occurrences with given replacement string

- and ***search*** on strings, eg.:
  - ***strstr*** : find the first occurrence of a string
  - ***strpos*** : find the position of the last occurrence of a substring in a string
  - ***strcmp*** : string comparison

The PHP language syntax

I. Operations and functions on strings

- string ***trim*** (string $string [, string $charset])

```php
<?php
 $text   = "\t\tThese are a few words :) ...  ";
 echo $text; //"             These are a few words :) ..."

 $trimmed = trim($text);
 echo $trimmed; //"These are a few words :) ..."

 $trimmed = trim($text, " \t.");
 echo $trimmed; //"These are a few words :)"
?>
```

The PHP language syntax

I. Operations on strings:

- string ***substr*** (string $string , int $start [, int $length])

```php
<?php
  echo substr('abcdef', 1);     // bcdef
  echo substr('abcdef', 1, 3);  // bcd
  echo substr('abcdef', 0, 4);  // abcd
  echo substr('abcdef', 0, 8);  // abcdef
  echo substr('abcdef', -1, 1); // f
?>
```

- int ***substr_count*** (string $string , string $search_substr)

```php
<?php
  $text = 'This is a test';
  echo strlen($text); // 14

  echo substr_count($text, 'is'); // 2
?>
```

The PHP language syntax

I. Operations on strings:

- string ***str_replace*** (string $search , string $replace , string $original_str)

```php
<?php
  // Provides: Hll Wrld f PHP
  $vowels = array("a", "e", "i", "o", "u", "A", "E", "I", "O", "U");
  $onlyconsonants = str_replace($vowels, "", "Hello World of PHP");

  // Provides: You should eat pizza, beer, and ice cream every day
  $phrase  = "You should eat fruits, vegetables, and fiber every day.";
  $healthy = array("fruits", "vegetables", "fiber");
  $yummy   = array("pizza", "beer", "ice cream");

  $newphrase = str_replace($healthy, $yummy, $phrase);
 ?>
```

The PHP language syntax

I. Operations and functions on strings

- string ***strstr*** (string $string , string $search)
  Returns the right part of $string which starts with the first occurence of $search.

```php
<?php
  $email  = 'name@example.com';
  $domain = strstr($email, '@');
  echo $domain; // prints @example.com
?>
```

- int ***strcmp*** (string $string1 , string $string2)
  Returns < 0 if str1 is less than str2; > 0 if str1 is greater than str2, 0 if they are equal, and -1 if they are different but equal in length.

```php
<?php
  $str1 = "bear";
  $str2 = "";
  echo strcmp($str2, $str2); //0
  echo strcmp($str1, $str2); // 4
  echo strcmp($str2, $str1); //-4
?>
```

The PHP language syntax

I. Operations and functions on strings

- int ***strpos*** (string $string , string $search)

```php
<?php
  $mystring = 'abc';
  $findme   = 'a';
  $pos = strpos($mystring, $findme);

 // Note our use of ===.  Simply == would not work as expected
 // because the position of 'a' was the 0th (first) character.
 if ($pos === false) {
  echo "The string '$findme' was not found in the string '$mystring'";
 } else {
  echo "The string '$findme' was found in the string '$mystring'";
  echo " and exists at position $pos";
 }
>
```

The PHP language syntax

Object oriented
programming (OOP)

The PHP language syntax

Object Oriented programming

- In OO programming, objects are data with properties (state) and methods (actions).

- Main concepts:

Class - A programmer-defined data type, which includes local functions as well as local data.

Object - An individual instance of the data structure defined by a class.

Member Variable - The variables defined inside a class. This data will be invisible to the outside of the class and can be accessed via member functions.

Member function - A function defined inside a class, used to access object data.

The PHP language syntax

Object Oriented programming

- In OO programming, objects are data with properties (state) and methods (actions).

- Main concepts (advanced):

Inheritance - A class can be defined by inheriting existing function of a parent class. The child class will inherit all or few member functions and variables of a parent class.

Polymorphism - When the same function can be used for different purposes or behaves in different ways depending on the number of arguments, on the caller (parent or child class), etc.

Overloading - A type of polymorphism in which some or all of operators have different implementations depending on the types of their arguments.

The PHP language syntax

PHP and Object Oriented programming

- In OO programming, objects are data with properties (state) and methods (actions).
- PHP supports object-oriented programming.
- Example: a software object modeling a car

Properties:	Methods:
car->name="Fiat"	car->start()
car->model=500	car->drive()
car->weight=850	car->break()
car->color="white"	

The PHP language syntax

PHP and Object Oriented programming

- Example (class definition):

```php
<?php
 class Car {
     public $colour;
     public $fuel;

     public function start() {
        if($fuel>0){
          echo "start!\n";
          return true;
        }else{
          echo "no fuel!\n";
          return false;
        }
     }
 }
?>
```

- Example (class use):

```php
<?php
    include('Car.php');

    $my_car = new Car();

    // fuel charge
    $my_car->fuel = 10;

   // start my_car
   $started=$my_car->start();
>
```

Exception Handling: try/catch

The PHP language syntax

PHP and Exception handling through *try/catch*

- Exception handling is used to change the normal flow of the code execution if a specified error (exceptional) condition occurs

- **Try/Catch** flow control:

  The try statement allows to define a block of code to be tested for errors while it is being executed.

  The catch statement allows you to define a block of code to be executed, if an error occurs in the try block.

```php
<?php
    try {
        //  Block of code to try
    }
    catch(Exception e) {
        //  Block of code to handle errors
    }
>
```

The PHP language syntax

PHP and Exception handling through *try/catch*

- Example

```php
<?php
try {
    // try to open the file example.txt
    $file = fopen("example.txt", "r");
    // try to read the file
    $content = fread($file, filesize("example.txt"));
    // try to close the file
    fclose($file);
} catch (Exception $e) {
    // print the error message if an exception is trown
    echo "Error: " . $e->getMessage();
}

?>
```

The PHP language syntax

PHP and Exception handling: the Exception object

```
class Exception implements Throwable {

    /* Properties */
    protected string $message = "";
    private string $string = "";
    protected int $code;
    protected string $file = "";
    protected int $line;
    private array $trace = [];
    private ?Throwable $previous = null;

    /* Methods */
    public __construct(string $message = "", int $code = 0, ?Throwable $previous = null)
    final public getMessage(): string
    final public getPrevious(): ?Throwable
    final public getCode(): int
    final public getFile(): string
    final public getLine(): int
    final public getTrace(): array
    final public getTraceAsString(): string
    public __toString(): string
    private __clone(): void
}
```

The PHP language syntax

PHP and Exceptions: the ***throw*** statement

- Example

```php
<?php
  //create function with an exception
  function checkNum($number) {
    if($number>1) {
      throw new Exception("Value must be 1 or below");
    }
    return true;
  }

  //trigger exception in a "try" block
  try {
    checkNum(2);
    echo 'If you see this, the number is 1 or below';
  }
  catch(Exception $e) {
    echo 'Message: ' .$e->getMessage();
  }
?>
```